THAKUR COLLEGE OF ENGINEERING & TECHNOLOGY
(Approved by AICTE, Govt. of Maharashita & Affiliated to University of Mumbai)
A - Block, Thakur Educational Campus, Shyamnayaran Thakur Marg,
Thakur Village, Kandivali (East), Mumbai - 400 101.

TCET/EXAM/FRM/IP-03/06 REVISION: B

IN-SEMESTER EXAMINATION-II

F.E./F.T. (Semester-I)

SUBJECT – Chemistry

Branch: E&TC, E&CS, IT, MECH, M&ME, AI&ML **Date: 22/12/2025**
Div.: ALL **Timing: 12:00 PM to 1:00 PM**
Duration: 60 Minutes **Maximum Marks: 20**

Instructions –

1. All questions are compulsory.
2. Assume suitable data wherever necessary and state the assumptions made.
3. Diagrams / sketches should be given wherever necessary.
4. Use of logarithmic table, drawing instruments and non-programmable calculators is permitted.
5. Figures to the right indicate full marks.

Q.1		Answer any 5 of the following questions	Marks	Course Outcomes	Learning Levels
	a.	Jerry observed a metal railing develops reddish-brown patches and begins to deteriorate. Identify the process and state their effects.	2	CO4	R
	b.	Silver ornaments left uncovered on a table for some days, still do not undergo corrosion. Locate the reason.	2	CO4	R
	c.	Illustrate the characteristics used to define the good fuel.	2	CO5	U
	d.	Compare High Calorific Value and Low Calorific Value in the context of their values.	2	CO5	U
	e.	Describe the term Refining of Petroleum.	2	CO5	U
	f.	Establish the connection between Gibbs Phase Rule and solder alloy systems?	2	CO6	A
	g.	Compute the degree of freedom of one component system where all the phases co-exist in equilibrium. What is your conclusion on the obtained value?	2	CO6	A
Q.2	a.	A metropolitan city facing the problem of corrosion of underground pipelines. Mr. Aniket as a chief engineer suggested the different ways to prevent this. He emphasized more on the protection of pipelines. What different methods would have been suggested? Explain in detail the method which being more economic and effective with suitable diagrams and reaction if any.	5	CO4	A
		OR			
	b.	A gas has the following composition by volume CO=8%, CH_4=50%, H_2=20%, N_2=3%, CO_2=3%, O_2=2%, C_2H_6=15%. Calculate the volume of air supplied per m^3 of this gas.	5	CO5	A

Q.3	a.	A coal having following composition by weight C=62%, O=30%, H=6%, S=1%, N=0.3% and ash=3.0%. Calculate GCV and NCV by Dulong and Pettit's formula.	5	CO5	A
	OR				
	b.	Sketch the graph of one component system. Based on the graph analyse the different areas and curves.	5	CO6	A